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Peer e-Feedback and ChatGPT-4o in EFL Writing: A Cognitive-Interpersonal Comparison Based on EFL Students

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ABSTRACT

Background: Although both peer and AI-generated feedback are increasingly used in EFL writing instruction, little is known about how they differ in terms of cognitive depth and interpersonal delivery. Existing studies often overlook the mechanisms through which feedback operates, limiting instructors' ability to design balanced feedback ecosystems.

Purpose: To provide a comparative analysis of peer and ChatGPT-4o feedback on undergraduate EFL academic writing, drawing on Hattie and Timperley's (2007) cognitive model and Hyland and Hyland's (2006) interpersonal feedback strategies. It aims to identify how each source targets task, process, and self-regulation levels, and how their rhetorical styles shape learner engagement.

Method: Thirty Iranian undergraduate EFL students participated in a qualitative classroom-based study in which each essay received both peer and ChatGPT-4o feedback. A total of 430 peer comments and 224 ChatGPT feedback units were coded deductively using validated analytical frameworks. Inter-rater reliability for 20% of the dataset yielded substantial agreement (κ = .82).

Results: Peer feedback was subjective, socially expressive, and frequently mitigated, with comments focusing on sentence-level issues but also including self-regulation prompts (15%) that encouraged reflection and decision-making. ChatGPT-4o provided predominantly task-level feedback (94%), characterized by structured, consistent, and objective guidance on grammar, organization, and coherence. However, its feedback exhibited minimal interpersonal variation and rarely promoted metacognitive engagement.

Conclusion: The findings indicate that peer and AI-generated feedback serve complementary pedagogical functions: AI offers technical accuracy and consistency, while peer feedback contributes emotional support and occasional reflective prompts. A hybrid feedback model that integrates both sources may therefore enhance the revision process in EFL writing instruction.

KEYWORDS

peer e-feedback; chatgpt-4o; academic writing; feedback comparison

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INTRODUCTION

Academic writing is a complex skill that often challenges students to meet desired standards (Kerman et al., 2022). A critical component of writing development is feedback, which plays a key role in enhancing writing skills and fostering self-awareness, motivation, and self-regulation (Hattie & Timperley, 2007; Zhang & Hyland, 2022). Feedback can take various forms, including teacher and peer feedback. While teacher feedback has traditionally been the primary approach, increasing workloads may limit its effectiveness and place pressure on instructors. As a result, peer feedback has emerged as a valuable alternative, offering students opportunities for collaboration and critical engagement in the writing process (Hattie & Timperley, 2007). Recent developments in large language models (LLMs) such as ChatGPT-4 have introduced a new avenue for providing writing feedback at scale, particularly in EFL and academic contexts (Shi & Aryadoust, 2024). When guided by teachers, AI can complement traditional feedback methods, enhancing their efficiency and adaptability to individual student needs (Farrokhnia et al., 2023).

This aligns with Hattie and Timperley's (2007) cognitive model, in which effective feedback facilitates learning at task, process, and self-regulation levels. At the same time, its ability to provide immediate and detailed language assistance is especially relevant from an interpersonal perspective (Hyland & Hyland, 2006), since supportive and timely feedback can reduce anxiety and sustain motivation for EFL students (Steiss et al., 2024). Despite promising findings, most recent studies on LLM-generated feedback have focused either on textual outcomes or general student satisfaction, with limited attention to the underlying structure and delivery of feedback itself (e.g., Alnemrat et al., 2025; Asadi et al., 2025). In particular, they often evaluate *what* feedback is provided but neglect *how* learners experience and respond to it cognitively and emotionally (Zhan & Yan, 2025). This creates a significant blind spot: without understanding the mechanisms and delivery modes of AI and peer feedback, instructors cannot make informed decisions about how to balance or sequence these sources in writing pedagogy.

To delve into these gaps, the present study applies two complementary theoretical frameworks: Hattie and Timperley's (2007) cognitive model and Hyland and Hyland's (2006) interpersonal model of feedback. These frameworks, when used in tandem, offer a more complete analytical lens. Hattie and Timperley's model enables us to trace the pedagogical *function* of feedback, does it target surface errors, strategic thinking, or self-regulated revision? Meanwhile, Hyland and Hyland's framework captures the rhetorical delivery of feedback, whether it motivates, supports, or alienates the learner through tone, hedging, or emotional cues.

The study aimed to answer the following questions:

RQ1: What types of feedback do peers and ChatGPT-4o provide, based on the task, process, and self-regulation levels described in Hattie and Timperley's (2007) model?
RQ2: How do peers and ChatGPT-4o differ in the way they deliver feedback, according to the interpersonal strategies outlined by Hyland and Hyland (2006)?
RQ3: What are the main similarities and differences between peer and ChatGPT-4o feedback in the context of EFL academic writing?

By integrating both models, we go beyond simple outcome comparisons and uncover the interaction between message and manner, that is, what is said, how it is said, and how these affect learning. In doing so, we respond to recent calls for more nuanced, classroom-grounded evaluations of LLM feedback in real-world contexts (Yu & Xie 2025). This study also aligns with the growing call for improving students' "feedback literacy," or their ability to interpret, evaluate, and apply AI-generated feedback critically and reflectively (Zhan & Yan, 2025).

THEORETICAL FRAMEWORK

Hattie and Timperley's (2007) Four-level Model of Feedback

This study uses Hattie and Timperley's (2007) four-level model of feedback as a conceptual foundation for comparing peer feedback with ChatGPT-generated feedback. The model views feedback not just as correction but as information that helps learners reduce the gap between their current performance and the goals of a task. What makes this model especially helpful is the way it breaks feedback down into four levels: task, process, self-regulation, and self, each reflecting a different kind of support that learners might receive.

Task-level feedback focuses on whether the work is correct, complete, or aligned with expectations. It addresses the content directly, often involving specific corrections or affirmations, and is most effective when it helps the learner understand how to improve accuracy (Hattie & Timperley, 2007, p. 90). This kind of feedback is common in both peer and AI-generated responses because it responds to the surface features of performance. For instance, a peer might comment, *"Your argument is unclear here, please rephrase,"* while ChatGPT might suggest, *"Use the past tense for consistency."* In fact, most written feedback in higher education remains task-focused (Lipsch-Wijnen & Dirkx, 2022).

Process-level feedback deals with the strategies or approaches used to complete a task. Rather than focusing on what is right or wrong, it supports the learner's ability to organize ideas, structure responses, or apply techniques (Hattie & Timperley, 2007, p. 92). A peer might suggest, *"You could structure this argument better by starting with a counterpoint,"* while ChatGPT could note, *"Try to use more connecting words to improve flow."* Such feedback helps learners transfer knowledge across tasks and contexts (Butler & Winne, 1995).

Self-regulation feedback is the most metacognitive of the four levels. It involves guidance that helps learners reflect on their goals, assess their strategies, and plan next steps—essentially, feedback that encourages self-monitoring and self-direction (Hattie & Timperley, 2007, p. 93). Though less frequently provided, this level has strong potential for supporting learner autonomy and long-term improvement. Hattie and Timperley argue that self-regulation is activated when feedback includes cues for self-assessment and error correction (p. 94). Mandouit and Hattie (2023) further support this, showing that learners value feedback that helps them identify "where to next," linking clearly to the feed-forward dimension of the model. Self-level feedback shifts the focus from the work to the individual, usually in the form of praise or personal evaluation. While such feedback may seem encouraging, it tends to have little impact on learning unless it is directly tied to performance (Hattie & Timperley, 2007, p. 96).

Hyland and Hyland's Interpersonal Model

Hyland and Hyland's interpersonal model focuses not on what feedback addresses, but on how it is communicated. They argue that feedback is both an instructional and interpersonal act that shapes the emotional and social dynamics of the writing process. Their model identifies several dimensions through which feedback varies rhetorically.

Orientation refers to the degree of control embedded in the feedback message. Directive comments give explicit instructions (e.g., *"Add a topic sentence"*), while facilitative comments invite learner agency and decision-making (e.g., *"Could a clearer topic sentence improve this paragraph?"*). In classroom contexts, facilitative forms are frequently preferred, particularly in EFL contexts where students may feel socially constrained or unsure about asserting authority (Yu et al., 2015). For instance, in our study one student wrote, *"You might consider adding an example here, it could clarify your point,"* instead of directly instructing a peer. This pattern reflects the politeness strategies often used to maintain solidarity and avoid confrontation in peer interactions.

Linguistic mitigation refers to the use of hedging or softening language that reduces the face-threatening nature of critical feedback. Expressions like *"maybe," "perhaps,"* or *"you could"* help maintain politeness while signaling tentativeness. Hyland (2010) noted that tutors frequently employed hedged phrases such as "you might want to…" when offering revisions, thereby balancing critique with respect. Similarly, Yu and Hu (2017) observed in their analysis of peer feedback that EFL students often used mitigated forms to protect group harmony. Comments like "Maybe this paragraph could use a clearer transition?" which cushion suggestions without undermining the peer relationship.

The final dimension concerns emotional tone, which can range from encouraging and supportive to neutral or impersonal. A supportive tone often includes praise and encouragement (e.g., «Great start! Your thesis is clear.»), while a neutral tone avoids emotion (e.g., «Your thesis is stated.»), and an impersonal tone may feel mechanical or detached (e.g., «Thesis: present.»). These stylistic choices influence how feedback is received by learners and whether it motivates or discourages further engagement with the writing task. In classroom discourse, affectively rich peer feedback has been shown to increase motivation, especially among novice writers. For instance, one peer in our study wrote, *"I loved your conclusion, it really ties everything together,"* enhancing the recipient's confidence and willingness to revise. In a nutshell, Hyland and Hyland's model provides an essential complement to Hattie and Timperley's by revealing how interpersonal dimensions shape the uptake and emotional reception of feedback. While Hattie and Timperley help identify the level and content of feedback, Hyland and Hyland explain how the delivery style influences its impact. This is particularly important in today's context, where learners increasingly receive feedback not just from teachers or peers, but also from AI tools like ChatGPT. Understanding both *what* is said and *how* it is said offers a more complete picture of the pedagogical affordances and limitations of each feedback source.

LITERATURE REVIEW

Peer Feedback in Writing Development

Peer feedback is considered an effective method for improving students' learning processes, particularly in writing development. feedback plays a crucial role in enhancing both cognitive and interactional processes (Li et al., 2024). Theoretical frameworks emphasize the significance of feedback not only as a means of correcting errors but also as a tool for fostering deeper cognitive engagement and promoting socio-cultural interaction. Hattie and Timperley (2007) highlight the multidimensional nature of feedback, demonstrating its impact at various cognitive levels—task, process, and self-regulation. This process allows students to internalize feedback, enabling them to self-regulate their learning and enhance their writing performance through sustained cognitive effort. In the classroom environment, feedback can take various forms such as teacher feedback and peer feedback. Peer feedback has emerged as an efficient alternative to teacher feedback because it is timely, dialogic, and frequent, reducing reliance on the teacher's limited availability (Cui et al., 2023; Noroozi et al., 2023). Studies show that peer review fosters deeper cognitive engagement and self-regulation, since students must critically analyze writing and reflect on their own texts (Yu & Hu, 2017). Additionally, peer feedback is usually less intimidating and more supportive than teacher feedback, enhancing motivation and confidence (Cui et al., 2023; Wu & Schunn, 2021).

Comparative Research on Peer and AI Feedback

Recent years have seen a growing interest in the role of artificial intelligence (AI), particularly large language models like ChatGPT, in academic writing and feedback. Several studies have compared the AI-generated feedback with that of traditional peer and teacher feedback (Banihashem et al., 2024; Chan et al., 2023; Escalante et al., 2023; Fang et al., 2023; Fleckenstein et al., 2023; Huang & Teng, 2025; Shi & Aryadoust, 2024; Steiss et al., 2024; Usher, 2025; Wang et al., 2023; Yoon et al., 2023; Yu & Xie, 2025; Zeevy-Solovey, 2024). Recent work has explored ChatGPT and other generative AI tools in education from different angles. For example, Zeevy-Solovey (2024) investigated how EFL learners perceived feedback from peers, ChatGPT, and their teacher, finding that while students appreciated AI for speed and clarity, teacher feedback remained the most trusted, with peer feedback valued mainly for surface-level issues. Fang et al. (2023) evaluated ChatGPT's grammatical error correction by comparing it with established tools across several languages. They found that ChatGPT often produced fluent text and successfully detected many errors, showing promise as a correction tool. However, the study also uncovered important weaknesses.

ChatGPT sometimes overcorrected sentences by changing text that was already correct, introduced inconsistencies by fixing an error in one place but missing it in another, and relied heavily on prompt wording, with small changes leading to very different results. These limitations suggest that while ChatGPT is strong in fluency and broad error detection, it lacks precision and consistency, and its approach is not grounded in a clear linguistic or theoretical framework. Building on these strands, Usher (2025) offered a more systematic comparison of AI, peer, and teacher assessments in higher education, employing a mixed-methods design with a standardized rubric and structured prompting. This study showed that AI chatbots provided more detailed guidance than peers, though human grading aligned more closely, and stands out for its methodological rigor compared to earlier explorations.

An important point in previous studies is the approach they use to examine feedback, which plays a crucial role in shaping their findings. We observed that, some studies adopt a cognitive lens, viewing feedback primarily as a scaffold that helps students close performance gaps (e.g., Zou et al., 2025). Others foreground interpersonal aspects, focusing on tone, mitigation, and affect (Jiang & Hyland, 2025; Mo & Crosthwaite, 2025). However, few studies consider both perspectives simultaneously, resulting in fragmented and often reductive conclusions. Moreover, their reliance on surface-level revision metrics limited insights into how students cognitively or emotionally processed that feedback. For example, Jiang and Hyland (2025) emphasized only the affective limitations of AI-generated responses, noting the absence of interpersonal engagement while not addressing the cognitive dimension. Another issue in the literature is that, although validated frameworks such as Hattie and Timperley's or Hyland and Hyland's exist, many researchers have not employed them when analyzing feedback. While many studies employ diverse tools, ranging from surveys and rubric-based scores to revision logs, few apply a shared theoretical lens to guide their analyses. As a result, findings often remain unidimensional, emphasizing either accuracy or learner preference, while overlooking the multifaceted nature of written feedback. For example, Yu and Xie (2025) and Barrot (2023) suggest that the absence of a structured theoretical framework might have hindered their ability to interpret their data at a deeper level.

This issue recurs across several studies (Banihashem et al., 2024; Fang et al., 2023; Fleckenstein et al., 2023; Huang et al., 2021; Huang & Teng, 2025; Steiss et al., 2024; Usher, 2025; Yoon et al., 2023; Zeevy-Solovey, 2024). For instance, Banihashem et al. (2024) compared ChatGPT and peer feedback on 74 students' argumentative essays. Their results showed that ChatGPT was highly effective in identifying issues related to grammar and structure, while peers were better at commenting on content and meaning. However, their broad coding categories were not linked to cognitive theory. Similarly, Escalante et al. (2023) conducted two studies to examine student preferences and outcomes when receiving ChatGPT or human tutor feedback. In the first study, both groups made comparable progress in writing skills. In the second, students expressed mixed preferences, some valued the interactive and personal nature of human feedback, while others appreciated the clarity and reliability of AI-generated comments. We see that their study offered valuable insights; however, they did not use any established coding scheme, nor did they employ any framework or theory-driven method in their analysis. In a similar way, Fleckenstein et al. (2023) used rubric scoring to assess feedback but did not adopt a multidimensional framework, limiting their analysis to surface-level accuracy. It is safe to say that, since many previous studies relied on ad hoc methods of analysis or lacked theoretical frameworks, they were unable to produce quantitative insights. Moreover, the absence of established coding schemes or consistent analytical criteria makes these studies difficult to replicate, highlighting a key methodological limitation that warrants further attention.

Another issue in the current literature is that, while tone and interpersonal delivery are frequently mentioned, they are seldom the subject of systematic analysis. Steiss et al. (2024) introduced a rater-based evaluation of tone, but their analysis lacked grounding in an interpersonal model. Similarly, Zou et al. (2025) noted the use of praise in AI responses but offered only broad descriptive insights. Moreover, the metacognitive dimension of feedback has also received limited attention. Zhan and Yan (2025) and Chen et al. (2025) observed that AI-generated feedback often results in passive uptake: students accept suggestions without questioning or reflecting on them. However, because most of these studies rely on ad hoc surveys or coding schemes, they lack the methodological precision to explore such patterns in depth. A further limitation is the imbalance in comparative research. Most existing studies focus on teacher versus ChatGPT feedback, while research directly comparing peer and AI-generated feedback remains scarce. This leaves important questions unaddressed, especially in peer-mediated contexts where feedback is shaped not only by instructional goals but also by identity, social dynamics, and politeness strategies, factors that are particularly salient in EFL settings (Hyland & Hyland, 2006; Yu & Hu, 2017).

To address these gaps, current study addresses the above limitations in the current literature in several ways. First, it integrates two established models, Hattie and Timperley's (2007) cognitive taxonomy and Hyland and Hyland's (2006) interpersonal framework, to enable a multidimensional analysis of feedback. This dual lens allows us to examine not only what feedback targets (task, process, self-regulation), but also how it is delivered (tone, orientation, mitigation). Finally, most prior work used simulated writing tasks or short interventions. This study, in contrast, draws on extended classroom data collected over a semester, offering a more realistic view of how feedback functions in practice.

METHOD

Design

This study employed a qualitative comparative design to investigate the nature of written feedback provided by peers and ChatGPT-4o on undergraduate EFL argumentative essays. The focus was on how each feedback source differed or overlapped in terms of cognitive depth and interpersonal style, based on two established theoretical models. The qualitative approach was chosen because it allows for in-depth analysis of the structure, tone, and purpose of feedback comments, which are often subtle and context-dependent. This approach is particularly suitable for comparative feedback studies, as it supports close reading and theory-informed interpretation of short written texts (Goldsmith, 2021; Kopec, 2023)

Participants

30 undergraduate EFL students from the University of Mazandaran participated in this study. The majority were female (22 females, 8 males), and most were between 19 and 21 years old. This was their first formal academic writing course, taken as part of their English language degree program. Their English proficiency was generally at the upper-intermediate level (CEFR B2), though some students were at the B1 or C1 level based on classroom performance and diagnostic writing tasks. At the beginning of the course, all participants were informed about the research purpose and procedures, and written informed consent was obtained, students voluntarily agreed to participate with the assurance of anonymity and confidentiality. All data were securely stored in password-protected digital folders. The teacher was the first author.

Instruments

To collect students' peer feedback, we used Google Drive (see Appendix) to create e-portfolios of students' assignments. For the current study, the teacher created a shared Google Drive folder titled «Academic Writing, 2023» and gave students access to it. Subsequently, individual student folders were generated within this main folder, each labeled with a student's name to function as their personal e-portfolio. Students were then directed to upload the various drafts of their essays into their respective created folders. Note that all of the students' writing assignments were then provided to ChatGPT-4o to obtain its feedback.

Data Collection Procedure

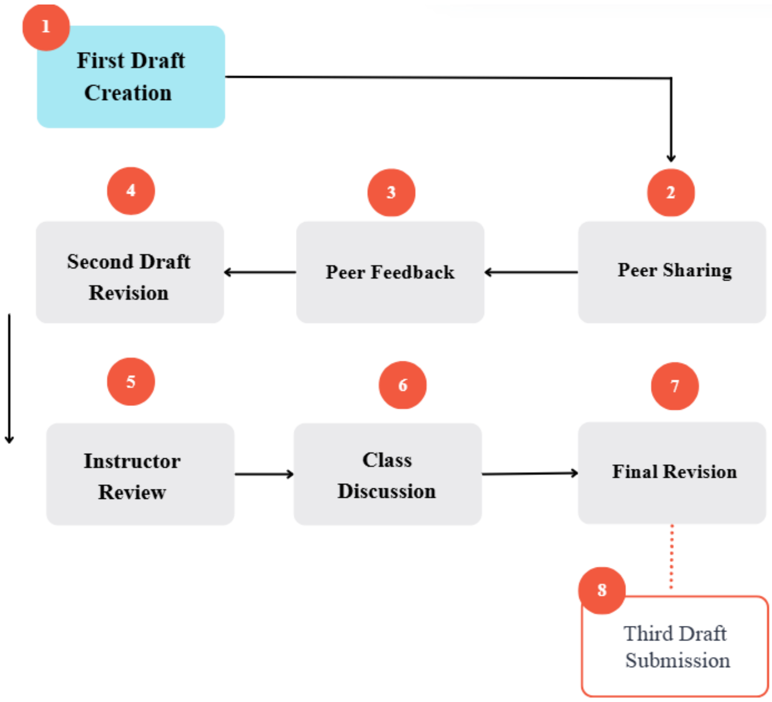
The study was conducted during a 15-week academic writing course in winter 2024, with weekly 90-minute sessions. 30 undergraduate EFL students participated, divided into six peer feedback groups. The Academic writing course spanned approximately four months (15 weeks), with weekly 90-minute sessions. The first session was dedicated to introducing the peer feedback process, during which the instructor formed six groups, each consisting of five students. From the second week onward, students were assigned three different essays, each following a three-stage cycle of feedback and revision. Students were trained how to write a paragraph for a few sessions and then they were gradually transit to a two-, and three-paragraph essay. The essay topics centered on everyday life issues and themes relevant to their field, such as «Advantages and Disadvantages of Studying in College,» «What is Metacognition,» «Motivating Factors for Undergraduate College Students».

Pedagogical Feedback Cycle

During the academic writing course, students worked on three essays, each progressing through three stages before the final version (see Figure 1). Initially, students wrote their first draft by reading various available online or in print materials, taking notes, outlining, and engaging in free writing. They then uploaded it to their e-portfolios and shared it with four other peers in their group via email. Peers provided feedback using track changes through Microsoft Word comments, and peer feedback was shared within the group. In the next stage, students revised their essays based on the peer feedback and uploaded the second drafts to their e-portfolios. Then, one student in each of the six groups, in turn, handed in the revised draft to the teacher. The teacher reviewed and provided comments on a total of 6 drafts, which were then collectively discussed in class. Importantly, this stage occurred only *after* students had exchanged and incorporated peer feedback into their second drafts. Thus, teacher input did not shape or interfere with the peer feedback process itself, since all peer comments were produced and acted upon before teacher review. Instead, the teacher comments and class discussions served to refine the second drafts further and support the transition to the third and final version. The third drafts were then uploaded to the students' e-portfolios, completing the pedagogical feedback cycle.

Figure 1

Pedagogical Feedback Cycle



ChatGPT Feedback

ChatGPT was not specifically trained or customized for its responses. That is, we let it be free in providing feedback and follow its approach freely.

Prompt Pilot Testing. Before selecting the final prompt for ChatGPT-4o, we tested several versions on a small sample of essays. We compared the responses for clarity, structure, and consistency. We chose a simple, open-ended prompt because it best reflected how students and teachers typically use ChatGPT, using straightforward instructions without advanced prompt design (Wang et al., 2024). Our goal was not to test ChatGPT's full potential through engineered prompts, but rather to observe how it naturally responds when asked to give feedback. Using the same prompt for all essays helped us capture this default behavior under realistic classroom conditions.

Prompt: "*I provide you with an essay written by undergraduate students. read thoroughly. Provide formative feedback, criticize, and comment on any aspects you think should be considered. Feel free to approach this task in your default way. The point is to figure out what you as ChatGPT 4o would do.*"

Data Analysis

This study used a qualitative content analysis approach guided by two established frameworks: Hattie and Timperley's (2007) model of feedback levels (task, process, and self-regulation) and Hyland and Hyland's (2006) interpersonal framework. We used principles from framework analysis to structure our coding process. This method is highly useful for applied educational research where the data is shaped by theory and the goal is to compare patterns across sources (Gale et al., 2013; Goldsmith, 2021; Stalmeijer et al., 2024). Although Hattie and Timperley (2007) identify four feedback levels, we did not include the "self-level" category in our coding scheme. Preliminary screening of our dataset confirmed that instances of purely self-level comments were minimal and often overlapped with interpersonal strategies (e.g., praise, mitigation) already captured by Hyland and Hyland's (2006) framework. Excluding this level avoided redundancy and ensured analytic focus on the three feedback dimensions most relevant to writing development.

Coding Procedure

Feedback from both sources (Peer and ChatGPT) was coded using a deductive approach. Each comment was treated as a separate unit of analysis. Peer feedback consisted of 430 individual comments. ChatGPT feedback (based on 30 essays) was broken down into 224 identifiable feedback units to allow for comparison. To ensure consistency and clarity in analysis, all feedback comments were entered into a coding spreadsheet created by the researchers (see table1).

To enhance reliability, a second coder who was an expert in the field independently analyzed a 20% sample of the data, which is a commonly used proportion in qualitative research for inter-rater reliability checks (Cheung & Tai, 2021). Each coder independently assigned codes to the same comments

Table 1

Sample Entries from the Coding Spreadsheet Used in the Analysis

ID	Source	Original Comment	Hattie & Timperley Level	Hyland & Hyland Strategy
1	Peer	I think you wrote a great paragraph and your supporting sentences supported your main idea!	Task-Level	Facilitative + Mitigated
2	Peer	Isn't this your concluding sentence? I think you should put it at the end of your paragraph.	Self-Regulation	Facilitative + Questioning
3	ChatGPT	Your writing shows great potential. Try to elaborate more in the body paragraphs.	Process-Level	Directive + Encouraging
4	ChatGPT	Consider breaking this long sentence into two for clarity.	Task-Level	Directive + Neutral
5	Peer	Maybe using past perfect is more appropriate.	Task-Level	Facilitative + Hedged
6	ChatGPT	The essay effectively contrasts emotions before and after college. Try to support it with more examples.	Process-Level	Directive + Neutral

using the predetermined frameworks. Their coded results were entered into a matrix comparing agreements and disagreements across categories. Cohen's Kappa was then calculated through SPSS 26 to account for agreement occurring by chance (McHugh, 2012). The agreement level between the coders yielded a Cohen's Kappa value of 0.82, indicating substantial to almost perfect agreement (McHugh, 2012).

RESULTS

Types of Feedback Peers and ChatGPT-4o Provide at the Task, Process, and Self-Regulation Levels

Cognitive Dimensions of Peer Feedback

A large share of peer comments aligned with task-level feedback, where students drew attention to grammar, vocabulary, and sentence mechanics. Rather than simply stating rules, they often expressed suggestions with gentle phrasing. One student, noticing a grammar issue, wrote, *"Maybe using past perfect is more appropriate,"* while another commented, *"I think you should write 'the parents' instead of 'parents'."* These examples reflect a strong awareness of linguistic form, but also an intention to remain non-confrontational. Phrases like *"I feel this part is extra,"* or *"There's something missing here, maybe an article,"* showed both attention to detail and a reliance on hedging to soften the impact of correction.

Moving beyond surface issues, some students provided process-level feedback, demonstrating awareness of coherence and structural quality. For example, one student remarked, *"You need to rewrite your first sentence and also connect it to the second,"* identifying a lapse in logical flow. Another encouraged improved transitions: *"Try to use more linking words like 'in contrast' or 'moreover' to connect your ideas."* Though not always specific, these suggestions show that students were applying classroom concepts and thinking about the reader's experience. One particularly thoughtful comment stated, *"Your body paragraphs are okay, but the topic sentences don't clearly show the point of comparison."* This reflects attention to how arguments unfold, not just to individual sentences.

Only a few comments reached the level of self-regulation, which is the most metacognitive type of feedback in Hattie and Timperley's model. These comments encouraged the writer to reflect, reconsider, or make decisions on their own. One student wrote, *"You know what you want to say, so try to make it clearer in your own way,"* which shifted the responsibility back to the writer. Another offered, *"You can revise this based on what you want to emphasize,"* leaving space for personal choice in revision. While these types of comments were less frequent, they suggested that some students were beginning to think of revision as more than just fixing errors, it was a flexible and reflective process.

Misaligned or Ineffective Peer Feedback

While many peer comments reflected effort and care, not all feedback was effective or aligned with the writer's needs. Some comments were too vague to guide revision meaningfully. For instance, *"Maybe try to change this part a little bit?"* did not specify what needed changing or why, leaving the writer without clear direction. Others offered suggestions based on instinct rather than rule-based understanding, such as *"You should use 'would' here because it sounds better,"* which, although well-meaning, could lead to confusion about grammatical accuracy.

In several cases, praise was used as a substitute for feedback. Comments like *"Great job! I really liked it,"* acknowledged the effort but lacked any specific reference to content, structure, or language. These kinds of remarks, while emotionally supportive, did little to contribute to the writer's revision process.

There were also moments of misunderstanding between the peer and the writer. One student wrote, *"I think you are trying to compare two ideas, but it's not clear which one you prefer,"* even though the original task was explanatory, not argumentative. This misalignment between task type and peer interpretation points to a deeper issue—how students perceive the purpose of the writing and how that perception shapes the feedback they give.

In some cases, the effort to be kind or indirect actually weakened the message. For example, *"Umm maybe I'm wrong but like... could this sentence be kind of too long or something?"* reflects a high degree of hedging, making it difficult for the writer to identify what specifically needs revision. Similarly, comments such as *"This reminds me of my cousin's story, haha 😅,"* while friendly and engaging, diverted attention from the task and provided no usable feedback.

These examples illustrate a central challenge in peer interaction: balancing emotional sensitivity with instructional clarity. For developing writers, such feedback can shape not only what they revise, but also how confident they feel doing so. When feedback lacks clarity or misinterprets intent, it can delay progress or create confusion - especially in early academic writing contexts where genre awareness and revision strategies are still forming.

Cognitive Dimensions of ChatGPT Feedback

ChatGPT's comments were primarily situated at the task and process levels. Much like a trained instructor, it routinely flagged sentence-level clarity, grammar, and phrasing. In one response, it advised, *"Consider breaking this sentence into two to improve readability,"* and in another, *"Avoid starting sentences with 'And' in formal writing."* These suggestions were direct and rule-based, offering immediate fixes without hedging or uncertainty.

At the process level, ChatGPT provided structural feedback, often noting whether topic sentences were clear or transitions smooth. It stated, *"The introduction sets up the topic well, but the thesis could be more specific,"* and *"The conclusion summarizes your main points but doesn't reflect back on the opening."* While this showed a grasp of essay architecture, the feedback was often generic. It followed a recognizable pattern, general praise, followed by issue identification, and then a recommendation. Though effective, it lacked the contextual nuance of a human reader engaging with meaning and intention.

What was notably missing was feedback that encouraged self-regulation. ChatGPT seldom prompted the writer to make choices, reflect on meaning, or consider multiple revision paths. Instructions like *"Add a supporting example here,"* or *"Rephrase this for clarity"* were practical but one-directional. They did not engage the writer as an agent in their own revision process. For students still learning how to think critically about their writing, this could result in revisions that are technically accurate but conceptually shallow.

Moreover, while ChatGPT was effective in pointing out errors, it occasionally failed to address meaning-level or rhetorical misalignments, especially when essays lacked clarity in purpose or tone. In several instances, it overlooked ambiguities in argument structure or misinterpreted an essay's intent due to its reliance on surface cues. This suggests that although ChatGPT can accelerate revision, it is less equipped to guide students through conceptual refinement, which is essential in academic writing development.

Ignoring Meaning-level Issues

Although ChatGPT-4o consistently provided accurate and organized feedback on grammar, structure, and style, it often failed to engage with deeper meaning-level or rhetorical issues in student essays. Its language-focused orientation allowed it to detect surface-level problems, but in doing so, it sometimes overlooked logical inconsistencies, vague reasoning, or conceptual confusion—areas that are critical for writing development.

In several cases, ChatGPT addressed surface phrasing while ignoring deeper problems of coherence or argument. For example, in one essay, a student wrote: *"Studying in college is very good because it helps you not be a loser."* ChatGPT responded, *"Consider using more formal language. 'Loser' may not be appropriate in academic writing."* While this advice was technically accurate, it missed the larger issue—the lack of elaboration and weak argumentative framing behind the statement.

Similarly, in the sentence *"Motivation is when students do hard and never stop until they reach happiness in their exam,"* ChatGPT flagged the phrase *"do hard"* and suggested replacing it with *"work hard"*. However, it ignored the vague and circular nature of the definition, as well as the problematic goal of achieving "happiness" through exams—an issue more semantic than syntactic.

There were also instances where ChatGPT applied generic praise to weak content. In one paragraph that stated, *"College give you skills. You will have better life and also job and marry,"* ChatGPT responded: *"This paragraph introduces a clear idea. Consider expanding on 'skills' with specific examples."* While encouraging elaboration, the feedback failed to address the overly simplistic reasoning, repetitive phrasing, and culturally narrow assumptions embedded in the sentence. These examples point to a recurring limitation in ChatGPT's current feedback design: its ability to detect and respond to language issues does not always extend to conceptual clarity, argument strength, or writer intent. As a result, students may receive grammatically accurate but rhetorically uncritical feedback, making it easier to revise form than to engage in deeper reflection or restructuring.

ChatGPT-4o and Peer Feedback Exhibit Distinct Interpersonal Delivery Patterns

Interpersonal Features of Peer Feedback

Interpersonally, peer comments often reflected a facilitative tone, rich in praise and emotionally supportive language. Many students began their comments with encouragement before offering critique. Comments like *"Well done dear 👍,"* or *"Nice paragraph 👏,"* were frequently used to acknowledge the writer's effort and soften the delivery of feedback. Even when addressing problems, students were careful with their language. A comment like, *"Just a suggestion, but I feel this sentence could be clearer,"* shows both attention to quality and a desire to avoid sounding harsh.

This interpersonal tone often shaped how feedback was received and possibly acted upon. Praise and emojis helped create a low-stress environment, potentially encouraging students to remain open to critique. For example, *"Maryam, I think starting with a dictionary definition was a smart move,"* not only offered a positive evaluation but also personalized the feedback, reinforcing peer-to-peer trust. Similarly, phrases like *"I really enjoyed your ideas 😊"* or *"It was perfect, just this one small part needs work"* reflected a culture of encouragement, which may support emotional investment in revision.

However, the emphasis on preserving harmony occasionally diluted the feedback's instructional value. Vague comments like *"I liked your conclusion"* or *"This feels vague and odd"* lacked specificity and were unlikely to lead to meaningful revision. Yet even these comments served a social function—helping students maintain connection and save face. This raises a key consideration: interpersonal strategies such as praise, hedging, and emoji use may increase feedback uptake by making students feel supported, but they may also limit revision quality when clarity is sacrificed for politeness.

From a developmental perspective, this balance between clarity and care is significant. Students in this setting were learning not only how to give feedback but how to manage emotional and relational dynamics while doing so. These early experiences in giving constructive yet sensitive peer feedback may contribute to long-term writing development by fostering reflection, dialogue, and revision ownership - even when the feedback itself isn't always precise.

Interpersonal Features of ChatGPT Feedback

ChatGPT's interpersonal stance was markedly different from that of peer feedback. Its tone was consistently formal, professional, and emotionally neutral. The system avoided personal pronouns, affective language, or any relational framing. Unlike peers who often began with phrases like *"Great job"* or *"Well done,"* ChatGPT opened its responses with statements such as, *"This essay has a clear structure but lacks elaboration in some parts."* These comments were informative but detached, offering little emotional cushioning.

Although this authoritative stance ensured clarity, it may feel impersonal or even cold to students, especially in early writing stages. There was no hedging (*"maybe," "perhaps"*) or affective language to ease the emotional impact of critique. Every comment was structured as a directive, leaving little room for dialogue or empathy. From Hyland and Hyland's perspective, ChatGPT operates purely in a directive mode, focused on task execution, not relationship-building.

While this authoritative tone helped ensure clarity and efficiency, it may have felt impersonal or even cold—especially for novice writers in the early stages of academic writing development. The lack of hedging (*"maybe," "it seems," "you might consider"*) or social softeners meant that all suggestions came across as final and unquestionable. Every comment was phrased as a directive or statement of fact, leaving minimal space for dialogue or negotiation. From Hyland and Hyland's (2006) perspective, ChatGPT's approach aligns with a purely directive mode, where feedback focuses on what needs to be done rather than how it might be discussed or co-constructed.

This impersonal format may affect how students interpret and act upon feedback. For some, the clarity and structure may be reassuring, offering confidence in what needs revision. But for others—especially those still building their identity as writers—the absence of warmth, praise, or encouragement may limit emotional engagement and reduce motivation to revise deeply. Without interpersonal cues, the feedback becomes more transactional than relational, potentially narrowing the revision process to surface-level corrections.

Key Similarities and Differences Emerge Between Peer and ChatGPT-4o Feedback in EFL Academic Writing

To illustrate the distribution of feedback types more clearly see Table 2 as it summarizes the proportion of each feedback level observed in peer and ChatGPT-4o comments. Note that Although ChatGPT feedback was originally generated for 30 essays, its responses were dissected into individual feedback units for analysis, mirroring the unit-level coding used for peer comments. This approach ensures the comparison presented in Table 2 reflects the actual distribution and nature of feedback.

Table 2

Comparison of Peer and ChatGPT Feedback across Cognitive Levels and Interpersonal Strategies (Based on Unit-Level Coding of Individual Feedback Points)

Dimension	Peer Feedback (n = 430)	ChatGPT Feedback (n = 224)
Task-Level	55%	60%
Process-Level	30%	38%
Self-Regulation	15%	2%
Facilitative Tone	65%	15%
Directive Tone	35%	85%
Mitigated/Hedged Language	58%	10%
Emotionally Expressive	40%	2%
Neutral/Formal Tone	10%	98%

Taken together, the peer and ChatGPT feedback revealed two distinct but complementary approaches to supporting student writing. While ChatGPT offered consistent, detailed, and technically accurate feedback, especially on grammar and structure, it largely remained at the task and process levels and lacked the nuance of reflection or personalization. On the other side, peer feedback was more variable in quality, but often rich in interpersonal tone and occasionally engaged with deeper aspects of content, meaning, and writer intention.

Peers brought emotional support, relational awareness, and at times a flexible understanding of context, often using praise, hedging, or reflective prompts to shape their responses. Their comments were socially sensitive and sometimes messy, but they revealed an effort to connect with the writer as a person. ChatGPT, on the other hand, maintained a professional, impersonal stance that emphasized efficiency and clarity, but offered little room for dialogue or independent decision-making.

DISCUSSION

The findings of this study revealed notable contrasts and overlaps between the two sources, particularly in terms of cognitive depth, tone, and delivery style. Peer feedback was detailed, subjective, and frequently focused on sentence-level issues, which aligns with the task level in Hattie and Timperley's feedback model. These comments typically addressed grammar, vocabulary, and phrasing, reflecting a surface-level focus, ChatGPT-4o, in contrast, offered structured and consistent feedback. Its responses emphasized essay structure, coherence, and grammatical accuracy, operating mainly at the task and process levels. These findings align with previous studies (Banihashem et al., 2024; Yu & Hu, 2017; Zeevy-Solovey, 2024). However, approximately 18% of peer comments extended beyond task-level, targeting argument building, planning, or paragraph flow, indicators of process and self-regulation levels. This demonstrates that, despite a tendency toward surface correction, some students engaged in deeper cognitive work when prompted by peer interaction.

In this study, peer feedback also showed variation in depth and clarity. This inconsistency, mirrors findings by Shi and Aryadoust (2024), who reported that peer feedback can range widely depending on students' proficiency and confidence. Peer comments also addressed the process level feedback by focusing on the logical development and thematic relevance of arguments, showing sensitivity to whether ideas were thematically appropriate. This supports Escalante et al. (2023), who observed that human feedback is more likely to address content appropriateness and meaning-making. Nevertheless, the variability in peer feedback underscores the need for structured training. Without explicit instruction in providing evidence-based and cognitively rich comments, students risk defaulting to vague praise or surface-level critique. However, only 4.2% of ChatGPT comments in our study involved process-level guidance, and fewer than 2% included self-regulation prompts.

These findings reveal a significant pedagogical gap, that is, ChatGPT may facilitate surface-level correction, but it does not support students in developing reflective revision habits or evaluative thinking. This confirms findings by Xie et al. (2025), who observed that students frequently accepted ChatGPT's suggestions without question, leading to passive uptake and limited metacognitive engagement. Teng (2024) similarly found that while students gained confidence using AI, they showed no significant improvement in self-monitoring or evaluative judgment. This lack of metacognitive prompting contradicts Hattie and Timperley's (2007) model, which positions self-regulation as the highest and most transformative level of feedback.

The consequences of this omission are profound. Students exposed primarily to ChatGPT risk becoming overly reliant on external correction and may not develop the internal dialogue necessary for independent revision. Moreover, this study showed that peers were cautious while providing feedback and this aligns with Banihashem et al. (2024) and Escalante et al. (2023) who also highlight this pattern of socially supportive peer interaction. In Iranian classrooms, where Hofstede's (2001) dimensions indicate high collectivism and uncertainty avoidance, emotionally neutral or overly direct feedback may be perceived as discouraging or overly critical. This mismatch can create dissonance, reducing students' willingness to engage with feedback and undermining its intended effect. Thus, pedagogically, the challenge is not only cognitive but cultural: educators must teach students to interpret directive tone not as authoritarian command, but as academic input open to negotiation. To mitigate these issues, we propose a blended feedback model that draws on both sources: ChatGPT can offer immediate, structured input on grammar and cohesion, followed by peer review sessions focused on content, tone, and meaning-making. This recommendation aligns with Wang et al. (2023), who advocate for hybrid feedback systems.

In practical terms, a classroom-based implementation might follow a staged sequence: students use ChatGPT to revise surface-level issues in an initial draft, then receive peer feedback guided by rubrics that emphasize rhetorical clarity, argument development, and interpersonal tone. The teacher mediates both stages: highlighting conflicts, reinforcing deeper prompts, and explicitly addressing how to interpret directive versus facilitative cues. This pedagogical training must go beyond operational fluency and address critical interpretive skills, as what Carless and Boud (2018) refer to as feedback literacy. Only through such integrative training can we develop writers who are not only accurate, but also strategic, self-regulated, and responsive to feedback in all its forms.

Also, we should note that while Hattie and Timperley's (2007) model offers a clear structure for analyzing feedback, it was developed in school settings and may not fully capture the complexity of university-level EFL writing, where revision is more recursive and rhetorically driven. Likewise, Hyland and Hyland's (2006) interpersonal model, though valuable, was not designed with AI-mediated feedback in mind, limiting its applicability to machine-generated tone and relational cues. These limitations suggest a need for expanded models that address both academic discourse socialization and evolving digital feedback forms. However, this study aimed to pave the way for future framework-based and theory-driven analyses in feedback research, especially in the age of AI-assisted language learning, by demonstrating how established models can be applied and adapted to emerging feedback contexts.

CONCLUSION

Overall, peer responses in our study reflected all three cognitive levels, task, process, and occasionally self-regulation, and were often delivered with a personal and facilitative tone. ChatGPT, on the other hand, mostly stayed within the task and process levels and maintained a neutral, directive voice, showing little variation in interpersonal approach. This study's contribution lies in its integration of a dual-theoretical framework, Hattie and Timperley's cognitive model and Hyland and Hyland's interpersonal framework, applied to real classroom data using validated coding schemes. This approach provides a replicable method for systematically analyzing multidimensional feedback, particularly in hybrid learning environments. The findings of this study have several implications for EFL teachers, students, and teacher training programs. First, teachers can improve the feedback process by recognizing the differences between AI and peer feedback. When each source's strengths are clearly understood, teachers can assign feedback tasks accordingly, making the process more efficient and productive. To support this, teachers should train students to use AI for general and structural revisions, while managing their expectations about what AI can and cannot offer. Helping students understand the unique roles of peer and AI feedback promotes greater independence in revision, as they learn when to rely on each source based on the feedback focus. More importantly, instructors must be prepared to manage these hybrid ecosystems pedagogically. This includes teaching students how to interpret directive versus facilitative feedback, reflect on their own writing goals, and synthesize suggestions from multiple sources. Without such guidance, learners may become overly reliant on AI or misinterpret its tone. Teacher training programs can also play a key role by equipping teachers with practical strategies for using AI feedback tools. Training should also address how to coach students in evaluating the credibility, tone, and depth of feedback as what Carless and Boud (2018) call "feedback literacy", to ensure they develop as autonomous writers in AI-augmented classrooms.

Limitations

One possible limitation of this study lies in the use of a single, open-ended prompt to elicit feedback from ChatGPT-4o. However, this choice was intentional. Our aim was not to test the full potential of AI feedback under ideal prompting conditions, but to examine how ChatGPT responds in its default mode, a scenario that reflects how students or instructors are likely to engage with the tool in practice. It is also possible that the wording of the prompt itself, which included the instruction to criticize, influenced the directive and impersonal tone of ChatGPT's feedback. While this phrasing was chosen to elicit a natural, unfiltered response, it may have inadvertently shaped the style of feedback produced. Other limitations include the homogeneity of the participant sample, limited to Iranian undergraduate EFL learners, which restricts generalizability across cultures and proficiency levels. Cultural norms around politeness and collectivism may have influenced the interpersonal tone of peer feedback. Moreover, the absence of revision tracking or learner perception data limits our ability to determine how feedback was received or applied. Finally, the qualitative nature of the analysis introduced a degree of subjectivity, and although mitigated through double-coding and the use of validated schemes, the possibility of confirmation bias cannot be fully eliminated.

Future Research

Future studies could explore how different types of prompts affect ChatGPT's feedback, particularly in relation to self-regulation and interpersonal tone. Comparing tailored versus default prompting strategies may offer further insight into how AI feedback can be optimized for classroom use. Another direction is exploring how students at different proficiency levels respond to AI feedback. Experimental designs that measure actual revision uptake or learning gains across feedback types would help establish causal links between source, tone, and learning outcomes. Finally, longitudinal studies could track how students improve over time when using both peer and AI feedback, identifying which type supports specific writing skills most effectively. Integrating real-time classroom data, prompt engineering experiments, and mixed-method feedback literacy interventions would further advance our understanding of AI-mediated feedback in diverse educational contexts.

DECLARATION OF COMPETING INTEREST

None declared.

AUTHOR CONTRIBUTION

Hossein Bozorgian: conceptualization; methodology; data curation; validation; writing – review & editing; supervision.

Hossein Rahimi: conceptualization; formal analysis; investigation; data curation; writing – original draft preparation.

REFERENCES

Alnemrat, A., Aldamen, H., Almashour, M., Al-Deaibes, M., & AlSharefeen, R. (2025). AI vs. teacher feedback on EFL argumentative writing: a quantitative study. *Frontiers in Education, 10*, Article 1614673. https://doi.org/10.3389/feduc.2025.1614673

Asadi, M., Ebadi, S., & Mohammadi, L. (2025). The impact of integrating ChatGPT with teachers' feedback on EFL writing skills. *Thinking Skills and Creativity, 56*, Article 101766. https://doi.org/10.1016/j.tsc.2025.101766

Banihashem, S.K., Kerman, N.T., Noroozi, O., Moon, J. & Drachsler, H. (2024). Feedback sources in essay writing: Peer-generated or AI-generated feedback? *International Journal of Educational Technology in Higher Education*, *21*(1). https://doi.org/10.1186/s41239-024-00455-4

Barrot, J. S. (2023). Using ChatGPT for second language writing: Pitfalls and potentials. *Assessing Writing*, *57*(1), Article 100745. https://doi.org/10.1016/j.asw.2023.100745

Butler, D. L., & Winne, P. H. (1995). Feedback and self-regulated learning: A theoretical synthesis. *Review of Educational Research, 65*(3), 245–281. https://doi.org/10.3102/00346543065003245

Carless, D., & Boud, D. (2018). The development of student feedback literacy: Enabling uptake of feedback. *Assessment & Evaluation in Higher Education*, *43*(8), 1315–1325. https://doi.org/10.1080/02602938.2018.1463354

Chan, C., Chen, J., Wang, W., Jiang, Y., Fang, T., Liu, X., & Song, Y. (2023). *ChatGPT evaluation on sentence-level relations: A focus on temporal, causal, and discourse relations* (arXiv preprint No. 2304.14827). https://doi.org/10.48550/arXiv.2304.14827

Cheung, K. K. C., & Tai, K. W. H. (2021). The use of intercoder reliability in qualitative interview data analysis in science education. *Research in Science & Technological Education, 41*(3), 1–21. https://doi.org/10.1080/02635143.2021.1993179

Chowdhury, M. F. (2015). Coding, sorting and sifting of qualitative data analysis: Debates and discussion. *Quality & Quantity*, *49*(3), 1135–1143. https://doi.org/10.1007/s11135-014-0039-2

Cui, G., Yuan, L., Ding, N., Yao, G., Zhu, W., Ni, Y., Xie, G., Liu, Z., & Sun, M. (2023). *UltraFeedback: Boosting language models with high-quality feedback* (arXiv preprint No. 2310.01377). https://doi.org/10.48550/arXiv.2310.01377

Cui, Y., & Schunn, C. D. (2024). Peer feedback that consistently supports learning to write and read: providing comments on meaning-level issues. *Assessment & Evaluation in Higher Education*, *49*(8), 1168–1181. https://doi.org/10.1080/02602938.2024.2364025

Dempere, J. M., Modugu, K. P., Hesham, A., & Ramasamy, L. K. (2023). The impact of ChatGPT on higher education. *Frontiers in Education*, *8,* 1206936. https://doi.org/10.3389/feduc.2023.1206936

Escalante, J., Pack, A., & Barrett, A. (2023). AI-generated feedback on writing: Insights into efficacy and ENL student preference. *International Journal of Educational Technology in Higher Education*, *20*(1). https://doi.org/10.1186/s41239-023-00425-2

Fang, T., Yang, S., Lan, K., Wong, D. F., Hu, J., Chao, L. S., & Zhang, Y. (2023). *Is ChatGPT a highly fluent grammatical error correction system? A comprehensive evaluation* (arXiv preprint No. 2304.01746). https://arxiv.org/abs/2304.01746

Farrokhnia, M., Banihashem, S. K., Noroozi, O., & Wals, A. (2023). A SWOT analysis of ChatGPT: Implications for educational practice and research. *Innovations in Education and Teaching International*, *61*(3), 1–15. https://doi.org/10.1080/14703297.2023.2195846

Fleckenstein, J., Liebenow, L. W., & Meyer, J. (2023). Automated feedback and writing: A multi-level meta-analysis of effects on students' performance. *Frontiers in Artificial Intelligence*, *6*. https://doi.org/10.3389/frai.2023.1162454

Gale, N. K., Heath, G., Cameron, E., Rashid, S., & Redwood, S. (2013). Using the framework method for the analysis of qualitative data in multi-disciplinary health research. *BMC Medical Research Methodology, 13*, Article 117. https://doi.org/10.1186/1471-2288-13-117

Goldsmith, L. J. (2021). Using framework analysis in applied qualitative research. *The Qualitative Report, 26*(6), 2061–2076. https://doi.org/10.46743/2160-3715/2021.5011

Hattie, J., & Timperley, H. (2007). The power of feedback. *Review of Educational Research*, *77*(1), 81–112. https://doi.org/10.3102/003465430298487

Hofstede, G. (2001). Culture's recent consequences: Using dimension scores in theory and research. *International Journal of Cross Cultural Management*, *1*(1), 11–17. https://doi.org/10.1177/147059580111002

Huang, J., & Teng, M. F. (2025). Peer feedback and ChatGPT-generated feedback on Japanese EFL students' engagement in a foreign language writing context. *Digital Applied Linguistics, 2,* 102469. https://doi.org/10.29140/dal.v2.102469

Huang, Y., Wang, J., & Chen, X. (2021). Automated writing evaluation in EFL contexts: Effects on writing quality and revision behavior. *Computer Assisted Language Learning*, *34*(3), 234–257. https://doi.org/10.1080/09588221.2020.1786712

Hyland, F. (2010). Future directions in feedback on second language writing: Overview and research agenda. *International Journal of English Studies, 10*(2), 171–182. https://doi.org/10.6018/ijes/2010/2/119251

Hyland, K., & Hyland, F. (2006). Feedback on second language students' writing. *Language Teaching*, *39*(2), 83–101. https://doi.org/10.1017/s0261444806003399

Jiang, F. & Hyland, K. (2025). Rhetorical distinctions: Comparing metadiscourse in essays by ChatGPT and students. *English for Specific Purposes*, *79*, 17–29. https://doi.org/10.1016/j.esp.2025.03.001

Johnson, C. I., Priest-Walker, H. A., Durlach, P. J., & Serge, S. R. (2012). The effect of feedback specificity in a virtual training environment. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, *56*(1), 1576–1580. https://doi.org/10.1177/1071181312561314

Kayaalp, F., Meral, E., & Başcı Namlı, Z. (2022). An analysis of the effect of writing-to-learn activities regarding students' academic achievement and self-regulation skills in writing. *Participatory Educational Research*, *9*(1), 324-348. https://doi.org/10.17275/per.22.18.9.1

Kerman, N. T., Noroozi, O., Banihashem, S. K., Karami, M., & Biemans, H. J. A. (2022). Online peer feedback patterns of success and failure in argumentative essay writing. *Interactive Learning Environments*, *32*(2), 1–13. https://doi.org/10.1080/10494820.2022.2093914

Khodadust, M. R. (2024). Politeness and corrective feedback: Immediate and delayed performance. *Iranian Journal of Language Teaching Research, 12*(2), 137–155. https://doi.org/10.30466/ijltr.2024.54354.2395

Kipp, M. (2024). From GPT-3.5 to GPT-4.o: A leap in AI's medical exam performance. *Information*, *15*(9), 543. https://doi.org/10.3390/info15090543

Kopec, A. (2023). Policy feedback & research methods: how qualitative research designs with marginalized groups inform theory. *International Journal of Qualitative Methods, 22*(10). https://doi.org/10.1177/16094069231217915

Li, C., Yang, Z., & Yang, Y. (2024). The impact of peer feedback on student learning effectiveness: A meta-analysis based on 39 experimental or quasi-experimental studies. In J. Gan, Y. Pan, J. Zhou, D. Liu, X. Song, & Z. Lu (Eds.), *Computer science and educational informatization: CSEI 2023* (Communications in Computer and Information Science, vol. 1899, pp. 42–52). Springer. https://doi.org/10.1007/978-981-99-9499-1_4

Lipsch-Wijnen, I., & Dirkx, K. (2022). A case study of the use of the Hattie and Timperley feedback model on written feedback in thesis examination in higher education. *Cogent Education*, *9*(1). https://doi.org/10.1080/2331186x.2022.2082089

Mandouit, L., & Hattie, J. (2023). Revisiting "The Power of Feedback" from the perspective of the learner. *Learning and Instruction*, *84*(84). https://doi.org/10.1016/j.learninstruc.2022.101718

McHugh, M. L. (2012). Interrater reliability: The Kappa Statistic. *Biochemia Medica*, *22*(3), 276–282. https://doi.org/10.11613/bm.2012.031

Mo, Z., & Crosthwaite, P. (2025). Exploring the affordances of generative AI large language models for stance and engagement in academic writing. *Journal of English for Academic Purposes*, *75*, 101499. https://doi.org/10.1016/j.jeap.2025.101499

Nakamura, S. (2018). Insights from studies on written corrective feedback: Implications for language pedagogy. *REFLections*, *22*, 89–102. https://doi.org/10.61508/refl.v22i0.112330

Noroozi, O., Alqassab, M., Kerman, N. T., Banihashem, S. K., & Panadero, E. (2024). Does perception mean learning? Insights from an online peer feedback setting. *Assessment & Evaluation in Higher Education*, *50*(1), 1–15. https://doi.org/10.1080/02602938.2024.2345669

Shi, H., & Aryadoust, V. (2024). A systematic review of AI-based automated written feedback research. *ReCALL*, *36*(2), 1–23. https://doi.org/10.1017/s0958344023000265

Stalmeijer, R. E., Klingberg, S., & Varpio, L. (2024). Using framework analysis methods for qualitative research: AMEE Guide No. 164. *Medical Teacher, 46*(5), 603–610. https://doi.org/10.1080/0142159X.2023.2259073

Steiss, J., Tate, T., Graham, S., Cruz, J., Hebert, M., Wang, J., Moon, Y., Tseng, W., Warschauer, M., & Olson, C. B. (2024). Comparing the quality of human and ChatGPT feedback of students' writing. *Learning and Instruction*, *91*, 101894–101894. https://doi.org/10.1016/j.learninstruc.2024.101894

Tang, T., Sha, J., Zhao, Y., Wang, S., Wang, Z., & Shen, S. (2024). Unveiling the efficacy of ChatGPT in evaluating critical thinking skills through peer feedback analysis: Leveraging existing classification criteria. *Thinking Skills and Creativity*, *53*, 101607–101607. https://doi.org/10.1016/j.tsc.2024.101607

Teng, M. F. (2024). "ChatGPT is the companion, not enemies": EFL learners' perceptions and experiences in using ChatGPT for feedback in writing. *Computers and Education: Artificial Intelligence, 7*, 100270. https://doi.org/10.1016/j.caeai.2024.100270

Usher, M. (2025). Generative AI vs. instructor vs. peer assessments: a comparison of grading and feedback in higher education. *Assessment & Evaluation in Higher Education*, *50*(6), 912-927. https://doi.org/10.1080/02602938.2025.2487495

Wang, J., Liang, Y., Meng, F., Sun, Z., Shi, H., Li, Z., Xu, J., Qu, J., & Zhou, J. (2023). Is ChatGPT a good NLG evaluator? A preliminary study. In Y. Dong, W. Xiao, L. Wang, F. Liu, & G. Carenini (Eds.), Proceedings of the 4th New Frontiers in Summarization Workshop (pp. 1–11). Association for Computational Linguistics. https://doi.org/10.18653/v1/2023.newsum-1.1

Wu, Y., & Schunn, C. D. (2021). The effects of providing and receiving peer feedback on writing performance and learning of secondary school students. *American Educational Research Journal, 58*(3), 492–526. https://doi.org/10.3102/0002831220945266

Xie, X., Zhang, L. J., & Wilson, A. J. (2025). Comparing ChatGPT Feedback and Peer Feedback in Shaping Students' Evaluative Judgement of Statistical Analysis: A Case Study. *Behavioral Sciences*, *15*(7), Article 884. https://doi.org/10.3390/bs15070884

Yoon, S.-Y., Miszoglad, E., & Pierce, L. R. (2023). *Evaluation of ChatGPT feedback on ELL writers' coherence and cohesion* (arXiv Preprint No. 2310.06505). arXiv. https://doi.org/10.48550/arXiv.2310.06505

Yu, H., & Xie, Q. (2025). Generative AI vs. teachers: Feedback quality, feedback Uptake, and revision. *Language Teaching Research Quarterly*, *47*, 113–137. https://doi.org/10.32038/ltrq.2025.47.07

Yu, S., & Hu, G. (2017). Understanding university students' peer feedback practices in EFL writing: Insights from a case study. *Assessing Writing*, *33*, 25–35. https://doi.org/10.1016/j.asw.2017.03.004

Yu, S., Lee, I., & Mak, P. (2015). Revisiting Chinese cultural issues in peer feedback in EFL writing: Insights from a multiple case study. *The Asia-Pacific Education Researcher, 25*(2), 295–304. https://doi.org/10.1007/s40299-015-0262-1

Zeevy-Solovey, O. (2024). Comparing peer, ChatGPT, and teacher corrective feedback in EFL writing: Students' perceptions and preferences. *Technology in Language Teaching & Learning, 6*(3), 1482. https://doi.org/10.29140/tltl.v6n3.1482

Zemach, D. E., & Rumisek, L. A. (2005). *Academic writing from paragraph to essay*. MacMillan.

Zhan, Y., & Yan, Z. (2025). Students' engagement with ChatGPT feedback: implications for student feedback literacy in the context of generative artificial intelligence. *Assessment & Evaluation in Higher Education*, 1–14. https://doi.org/10.1080/02602938.2025.2471821

Zhang, Z., & Hyland, K. (2022). Fostering student engagement with feedback: An integrated approach. *Assessing Writing*, *51*, 100586. https://doi.org/10.1016/j.asw.2021.100586

Zou, S., Guo, K., Wang, J., & Liu, Y. (2025). Investigating students' uptake of teacher- and ChatGPT-generated feedback in EFL writing: A comparison study. *Computer Assisted Language Learning*, 1–30. https://doi.org/10.1080/09588221.2024.2447279

APPENDIX

Examples of Google Drive and Google Docs Environments Used for Peer Feedback

Figure 2

An example of an e-portfolio using Google Drive

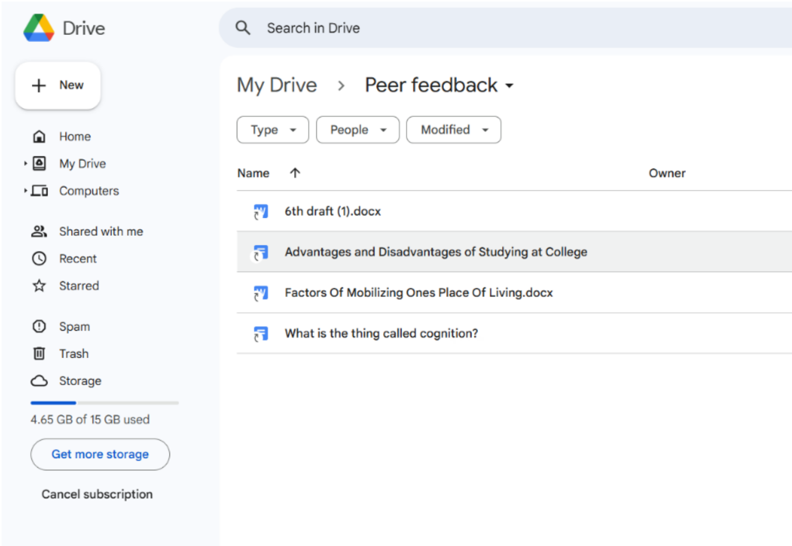


Figure 3

Collaborative environment of Google Docs with real-time peer feedback